Source Four
Assembly Guide

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Introduction

Welcome to the *Source Four Assembly Guide*.

Please note that the part numbers listed in this guide may differ from the parts required for your particular Source Four fixture. Part numbers for fixtures change occasionally as parts are replaced or upgraded.

To ensure that you are ordering the proper part for your specific fixture, please contact your ETC dealer, or ETC Customer Service for assistance.

Document Conventions

This document uses the following conventions to draw your attention to important information.



Note: *Notes are helpful hints and information that is supplemental to the main text.*



CAUTION: *A Caution statement indicates situations where there may be undefined or unwanted consequences of an action, potential for data loss or an equipment problem.*



WARNING: A Warning statement indicates situations where damage may occur, people may be harmed, or there are serious or dangerous consequences of an action.



WARNING: RISK OF ELECTRIC SHOCK! This warning statement indicates situations where there is a risk of electric shock.

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Please email comments about this manual to: TechComm@etcconnect.com.

Help from ETC Technical Services

If you are having difficulties, and your problem is not addressed by this document, try the ETC support website at support.etcconnect.com or the main ETC website at etcconnect.com. If none of these resources are sufficient, contact ETC Technical Services directly at one of the offices identified below. Emergency service is available from all ETC offices outside of normal business hours.

When calling for help, take these steps first:

- Prepare a detailed description of the problem
- Go near the equipment for troubleshooting
- Find your notification number if you have called in previously

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Basic Fixture

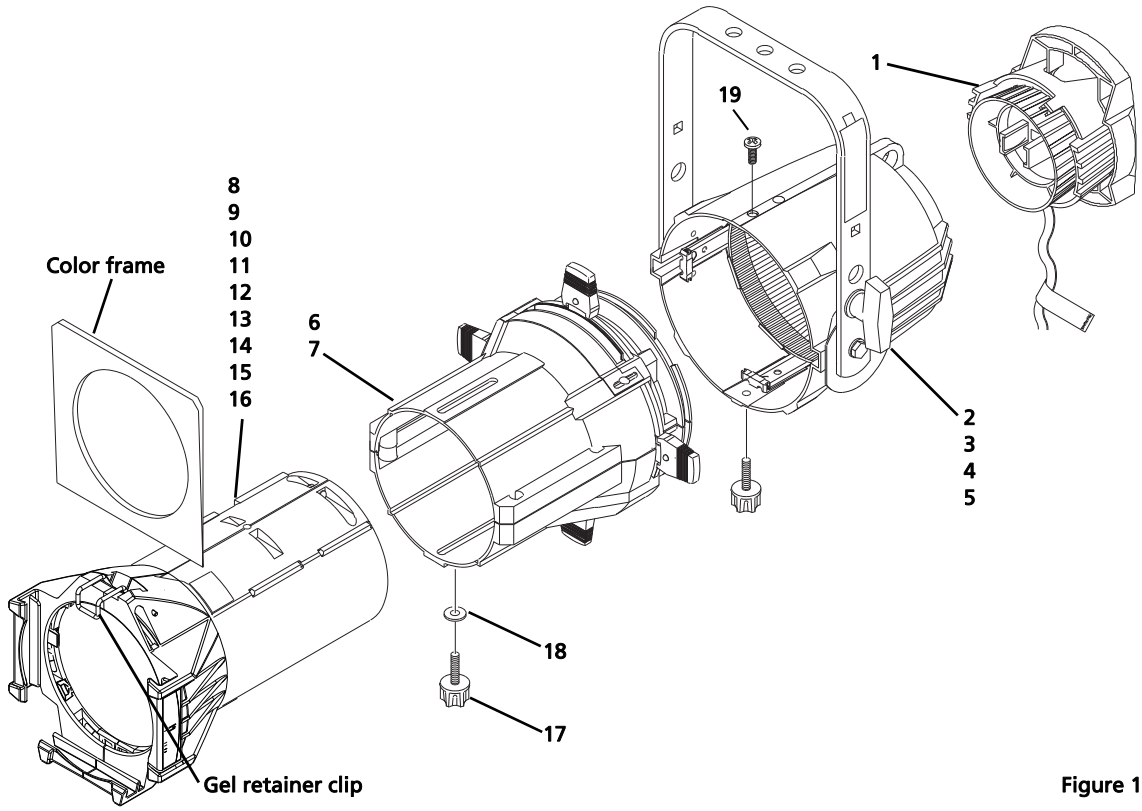


Figure 1

Reference Number	Part Number	Description	Quantity Required
1	7060A2008	Lamp burner assembly rated up to 750W, Black	1
2	7060A2011-K	Reflector housing assembly, single clutch, Black	1
3	7060A2011-K1	Reflector housing assembly, single clutch, White	1
4	7060A2020-K	Reflector housing assembly, double clutch, Black	1
5	7060A2020-K1	Reflector housing assembly, double clutch, White	1
6	7060A2012-K	Shutter barrel assembly, Black	1
7	7060A2012-K1	Shutter barrel assembly, White	1
8	7060A2000-K	5° lens tube, with knob	1
9	7060A2001-K	10° lens tube, with knob	1
10	7060A2050-K	14° lens tube, with knob	1
11	7060A2002-K	19° lens tube (6 x 16), with knob	1
12	7060A2003-K	26° lens tube (6 x 12), with knob	1
13	7060A2004-K	36° lens tube (6 x 9), with knob	1
14	7060A2005-K	50° lens tube (4.5 x 6), with knob	1
15	7060A2051-K	70° lens tube, with knob	1
16	7060A2052-K	90° lens tube, with knob	1
17	7060A4008-01	Knob set with male insert	2
18	HW5143	Washer, flat fiber	1
19	HW5197	Screw, 1/4-20 x 5/8, black zinc	1

Note: *Any part that is indicated as "Black" may be also ordered in white or silver. When ordering white, use the part number suffix -1, and for silver, use the part number suffix -5.*

Assembling basic fixture

1: Follow instructions under *Assembling lamp socket* on *page 6*.
2: Follow instructions under *Reflector Housing Assembly* on *page 9*.
3: Follow instructions under *Shuttle Barrel Assembly* on *page 12*.
4: Assemble Lens Tubes following the instructions for your applicable lens degree application.
5: Insert lens tube assembly (8-16) fully into the front barrel assembly (6-7). Align the mounting hole with the slot and secure with knob (17). See *Figure 1* on *page 3*.
6: Install reflector housing (2-5) onto front barrel assembly (6-7) and align the two mounting holes. Secure with screw (19) on top, and knob (17) on bottom. Tighten both securely. See *Figure 1* on *page 3*.
7: Install the lamp socket assembly (1) to the reflector housing (2-5) and secure with brass knurled screw on rear of lamp socket assembly.
8: Release the gel retaining clip by pushing it sideways while gently pulling backwards.
9: Insert the color frame.
10: Lock the clip by pushing sideways while gently pushing it forward.

Lamp Burner Assembly

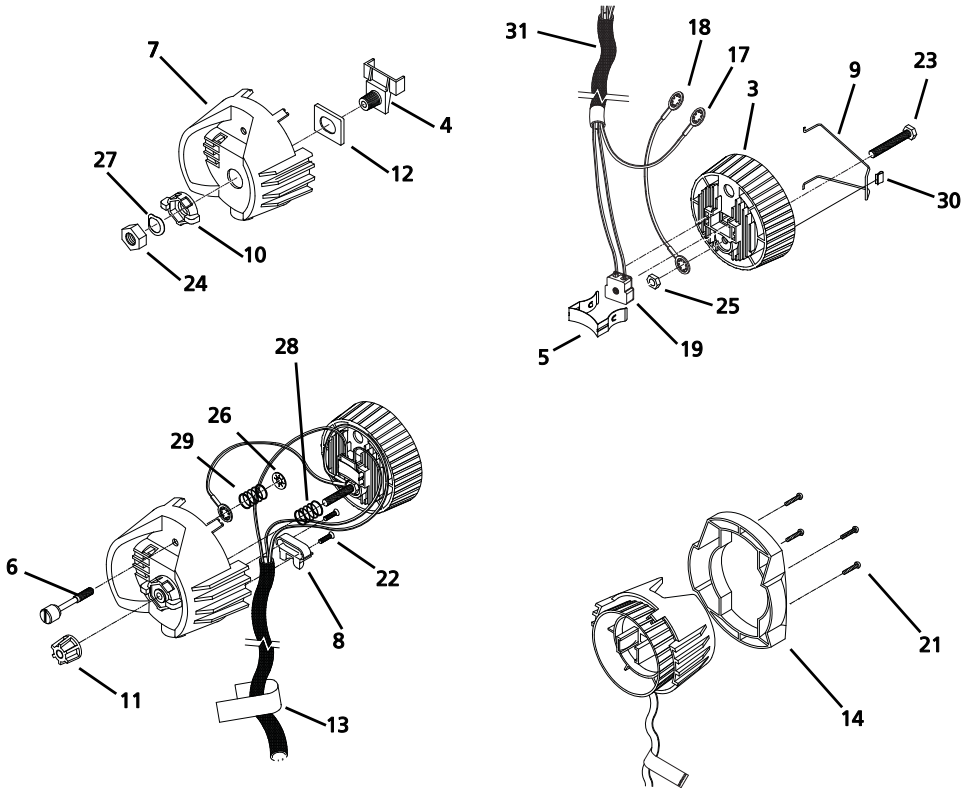


Figure 2

Reference Number	Part Number	Description	Quantity Required
1	7060A1033	Lamp retainer retrofit kit (Note: Does not include socket)	1
2	7060A2015	Light baffle retrofit kit Includes socket w/ mounting hardware and sleeve	1
3	7060A3007	Socket, light baffle casting (Note: For all related fixtures regardless of color)	1
4	7060A3011	Hub, index casting	1
5	7060A3012-0	Spring, lamp retainer, black oxide	1
6	7060A3025	Screw, 1/4-20 x 1.55 knurled head	1
7	7060A3055	Housing, socket, casting, Black	1
8	7060A3056	Clamp, strain relief, Black	1
9	7060A3085	Lamp retainer wire	1
10	7060A4007	Knob, X-Y, lamp set	1
11	7060A4008-02	Knob, Z, lamp set w/ female insert	1
12	7060A4011	Bushing cup	1
13	7060A4019	Label, Fixture Warning	1
14	7060A4037	Handle, Insulated Rear, Black	1
15	7060A4094	750W Handle label, Black	1
16	7060A4095	Label, 750W Handle, White	1
17	7060B7003	48" Source Four ground wire assembly W420 wire, 16 Awg, 200C/300V, green, UL 1180/CSA w/ J490T lugs	1
18	7060B7007	4" Source Four ground wire assembly W420 wire, 16 Awg, 200C/300V green, UL 1180/CSA w/ J490T lugs	1
19	7060B7019	S4 UL Socket Assy	1

Reference Number	Part Number	Description	Quantity Required
20	HW2125	Screw, 6-32 x 3/8" PhFHTC (lamp blocking screw not pictured)	1
21	HW2181	Screw, self tap, 6-32 x 5/8, black zinc	4
22	HW3103	Screw, 8-32 x 5/8 PhFHMS, black zinc	2
23	7060A3204	Bolt, 1/4-20 x 1 3/4", full thread	1
24	HW5123	Nut, hex, 9/16-18, black zinc	1
25	HW534	Nut, hex, 1/4-20, black zinc	1
26	HW746	Screw Retainer Flat 12-6-1	1
27	HW747	Washer, wave	1
28	HW748	Spring, compression CO720-072	1
29	HW749	Spring Ground	1
30	HW8209	Clip, Tinnerman	1
31	W6195	Sleeve, fiberglass 39", 5/16", Black	1
32	W689-1	Sleeve, fiberglass 39", 5/16", White	1

Note: *Any part that is indicated as "Black" may be also ordered in white or silver. When ordering white, use the part number suffix -1, and for silver, use the part number suffix -5.*

Assembling lamp socket

Tools Required:

- Open-end adjustable wrench or a 7/16" socket
- Needle-nose pliers
- #2 Phillips screwdriver

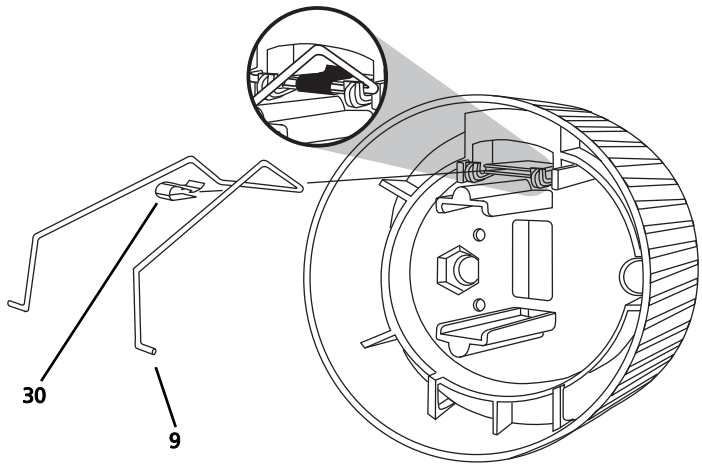


Figure 3

1: Set the crossbar of the retainer clip (9) under the two hooks on the clip bracket as shown in *Figure 3*.

2: Place the Tinnerman clip (30) over the retainer clip crossbar between the two hooks and press it down firmly until it snaps into place.

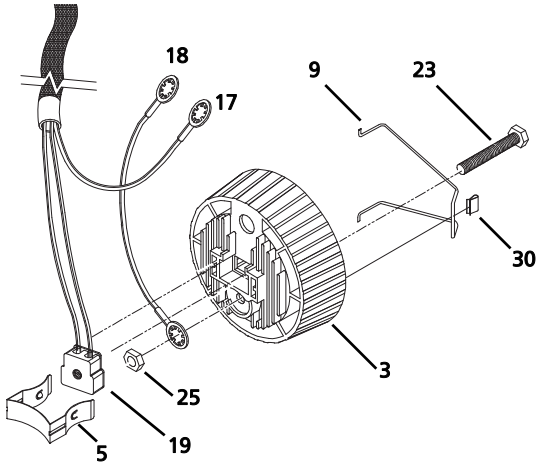


Figure 4

3: Insert the bolt (23) through the light baffle socket casting (3). See *Figure 4*.

4: Install the green ground wire assemblies (17 and 18) on the bolt (23) with the prongs on the crimped connectors toward the casting. Run both wires through the indent in the lip around the bolt hole. Secure with nut (25) and tighten securely (60 inch pounds).

5: Place the socket assembly (19) into the light baffle socket casting (3). Be sure it is well seated.

6: Install the lamp retainer spring (5). The lamp retainer spring secures the socket. Insert the spring one end at a time, making sure the rectangular slot in each side of the spring seats on the corresponding tab in the casting. See *Figure 4*.

Note: *If the spring does not seat correctly, coax it into place with a screwdriver or needle-nose pliers.*

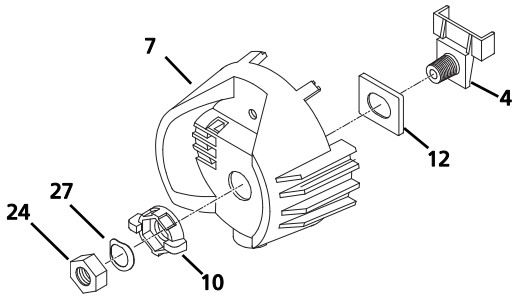


Figure 5

7: Install the bushing cup (12) into the housing socket casting (7) as shown in *Figure 5*. The cup should slide smoothly up and down, but not side to side.

8: Insert the threaded end of the index hub (4) through the holes in the bushing cup and the back of the housing socket casting (7).

9: Slide the X-Y knob (10) over the exposed index hub bolt (4), then insert the wave washer (27) on the bolt and secure with the 9/16 hex nut (24). Hand tighten the X-Y knob (10).

Note: *Install the wave washer with the upward curve toward the hex nut.*

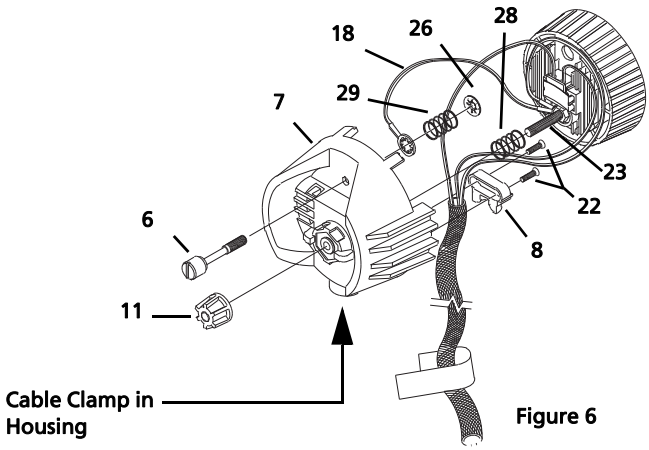


Figure 6

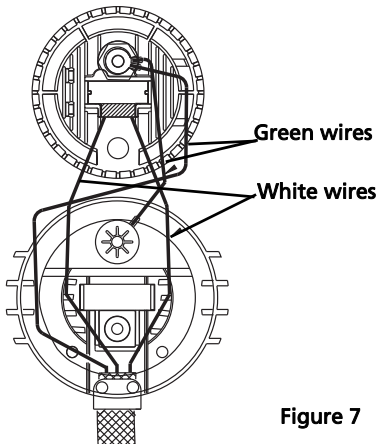


Figure 7

10: Insert the knurled head screw (6) through the housing socket (7) casting as shown in *Figure 6*.

11: Install the shorter green ground wire (18) onto the screw. The prongs on the crimped connector should be toward the casting.

12: Install the ground spring (29) onto the screw and secure it with the Southco flat retaining ring (26). Install the Southco ring with its prongs away from the casting.

Note: *Use pliers to straighten the Southco retaining ring (26) if it bends when you install it on the bolt.*

13: Lay the leads in the bottom half of the cable clamp (located in the housing socket casting [7]), making sure that the fiberglass sleeving extends slightly past the screw holes in the housing socket casting. Install new sleeving if necessary. Then, route the wires as shown in *Figure 7*.

CAUTION: *You must follow the wire routing diagram to ensure that the socket leads do not interfere with the lamp focus mechanism.*

14: Install the top half of the cable clamp (8) and secure it with the two screws (22). Tighten the screws alternately to ensure a solid connection.

Note: *To ensure that the clamp holds the cable tightly, flatten the cable, then fold over the sleeving before you install the top half of the clamp. Make sure the top edge of the cable clamp is even with the edge of the socket to prevent interference with lamp focus movement. Make sure sleeving is not pinched.*

15: Using the four screws (21), attach the handle (14) to the lamp socket assembly. See *Figure 2*.

16: Place the compression spring (28) on the protrusion on the inside of the index hub. See *Figure 6*.

17: Insert bolt (23) through spring (28) and through the index hub (4) of the housing socket (7), joining the two castings as shown in *Figure 6*. Make sure wires are not pinched between the two pieces.

18: Before proceeding, check again to make sure the wires are still positioned as indicated in *Figure 7*. Adjust if necessary.

19: Press the two castings together firmly so the bottom of the light baffle (3) sits on top of the cable clamp (8), then install Z lamp knob (11). Hand tighten the knob all the way to the right. See *Figure 6*.

CAUTION: *You must install Z knob as described above to ensure proper lamp focus travel.*

Reflector Housing Assembly

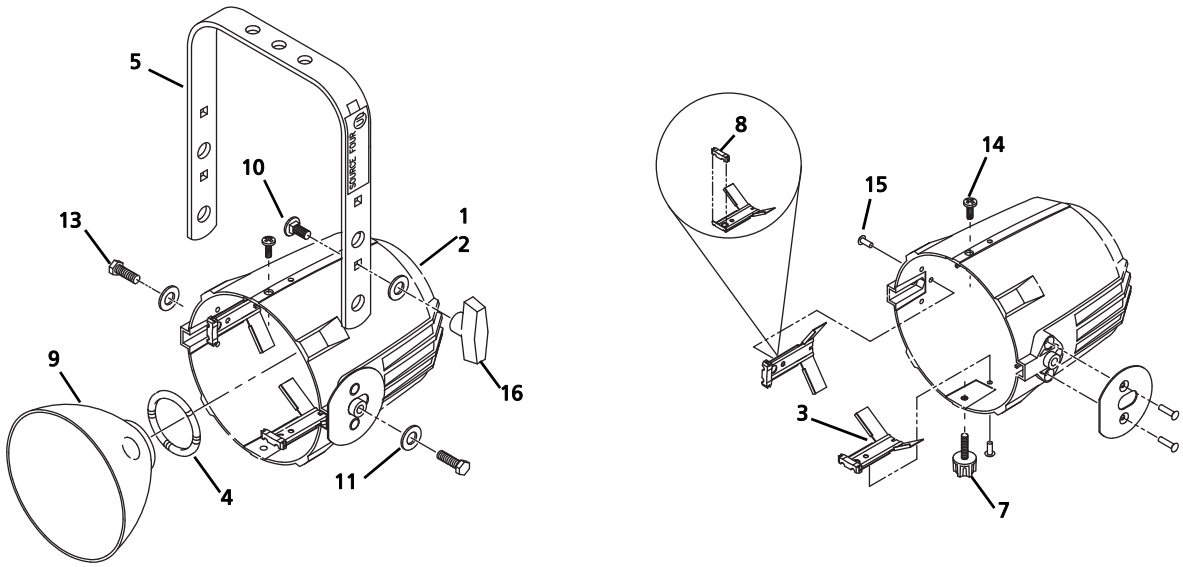


Figure 8 **Double-clutch reflector housing for 5°and 10° fixtures**

Reference Number	Part Number	Description	Quantity Required
1	7060A2013-K	Reflector housing rivet assembly, single clutch, Black	1
2	7060A2019-K	Reflector housing rivet assembly, double clutch, Black	1
3	7060A3006	Clip, reflector retainer	4
4	7060A3019	Spring, reflector support	1
5	7060A3058	Bracket, Yoke, Black	1
6	7060A3080	Light leak baffle, Black	2
7	7060A4008-01	Knob set with male insert	1
8	7060A4010	Bushing, gate	4
9	7060A4015	Reflector, molded glass, coated	1
10	HW5125	Bolt, carriage 5/16-18 x 3/4" LG	2
11	HW5126	Washer, flat, 5/16, black zinc	4
12	HW5127	Nut, hex, 5/16-18	2
13	HW5193	Bolt, hex, 5/16-18 x 3/4", black zinc	2
14	HW5197	Screw, PH, 1/4-20 x 5/8", black zinc	1
15	HW752	Rivet, machine, 3/16 x 31, oval head, black zinc	4
16	HW8144	Handle, yoke, knob, 5/16-15, Black	2

Note: *Any part that is indicated as "Black" may be also ordered in white or silver. When ordering white, use the part number suffix -1, and for silver, use the part number suffix -5.*

Removing reflector

Tools Required:

- Minimally padded work surface (cardboard, carpet, or rubber mat recommended)

WARNING: This procedure may crack or break the reflector. Always wear gloves, safety glasses, and a dust mask when performing this procedure.

1: Place the reflector housing face down on your work surface so that the concave reflector surface points downward.

2: Loosen the clutch and rotate the yoke so that it is perpendicular to the housing at roughly a 90° angle. Tighten the clutch.

3: Using the yoke as your handle, raise the housing assembly off of the work surface a few inches and then firmly tap the housing on the work surface. This should force the reflector out of its clips.

4: Carefully lift the housing to see if the reflector is released. If it is not, repeat step 3 with slightly more force.

Installing reflector

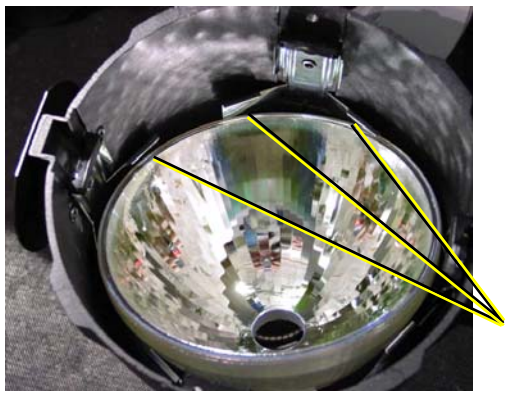
Note: *The following assumes all four reflector retainer clips (2) have already been riveted to the reflector housing casting (1) and the gate bushings (3) installed on the retainer clips. See Figure 8.*

1: Place the reflector housing casting (1-2) on a flat work surface with the large opening facing up.

2: Install the reflector support spring (4) in the circular opening at the base of the casting.

3: Insert the reflector (9) at an angle, under any three adjacent legs of the reflector's clips.

4: Gently press down on the opposite side of the reflector until it snaps into place under the remaining clips.



Step 3



Step 4

Note: *If the reflector does not snap in, turn the casting over. Gently pull on the side of the reflector that has not snapped in until the rest of the reflector slides into place.*

Cleaning reflector

WARNING: **Do not use ammonia-based or other harsh commercial cleaners. Clean reflector only as directed.**

Commercially available glass cleaning agents should be avoided as they may contain ammonia, other harsh chemical detergents or abrasive agents. These cleaners may damage the glass surface and the Anti-Reflective coatings. Do not immerse or soak the glass in any cleaning solution.

Remove dust with a blast of oil-free air or wipe with a clean, lint-free cloth. Isopropyl alcohol, distilled water or a 50%-50% mixture of each can be used to clean the glass surface.

Shuttle Barrel Assembly

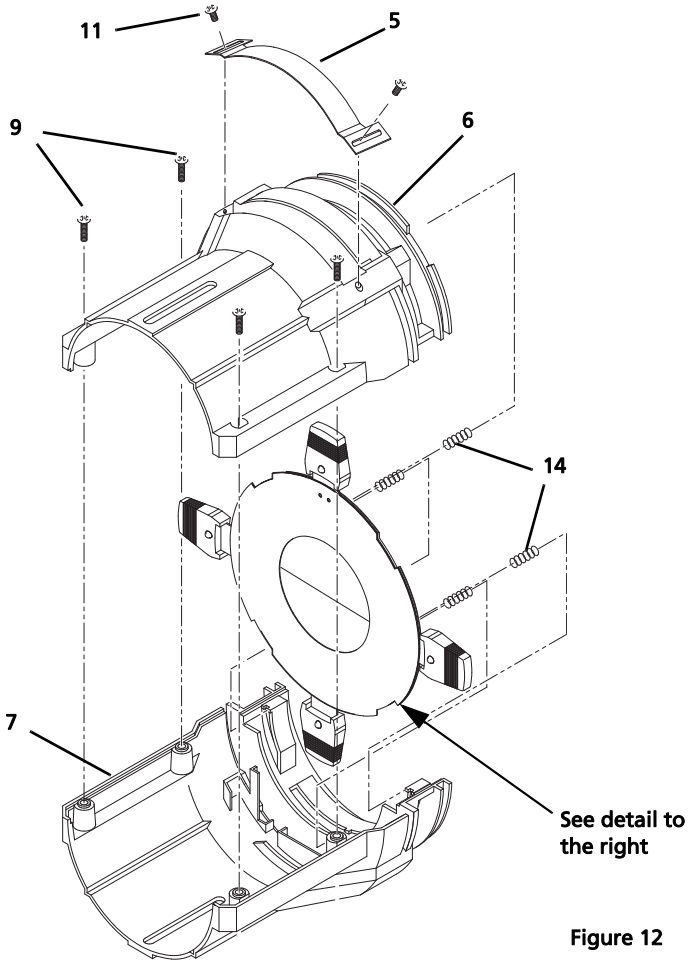


Figure 12

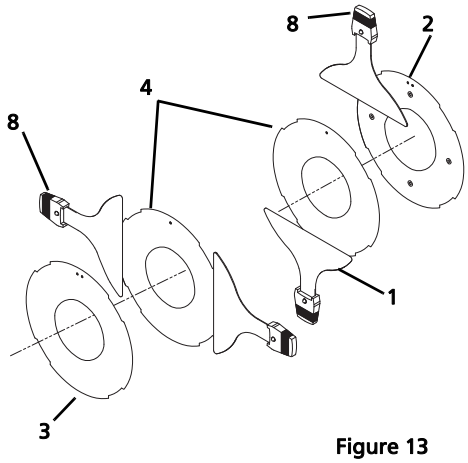


Figure 13

Shutter Assembly: The bottom divider plate (2) has four dimples punched into the surface; the top plate (3) has none. The middle divider plates (4) are noticeably thinner-gauge metal than the other two.

Reference Number	Part Number	Description	Quantity Required
1	7060A2025	Shutter blade assembly	4
2	7060A3001-01	Plate, divider with dimples (bottom)	1
3	7060A3001-02	Plate, divider (top)	1
4	7060A3003	Plate, gate (middle)	2
5	7060A3045	Cover, Iris Slot, Black	1
6	7060A3052	Shutter barrel, top casting, Black	1
7	7060A3053	Shutter barrel, bottom casting, Black	1
8	7060A4006	Shutter blade handle	4
9	HW3165	Screw, 8/32 x 1/2, PhPH Taptite Blz	4
10	HW232	Screw, 6-32 x 1/4" (Slot cover retention: pre 11/99)	2
11	HW372	Screw, 8-32 x 1/4" (begins 11/99)	2
12	HW370	Nut, Ny-lok, 8/32, black zinc	4
13	HW391	Screw, 8-32 x 3/4" PhPHMS, black zinc	4
14	HW754	Shutter spring	4

Note: *Any part that is indicated as "Black" may be also ordered in white or silver. When ordering white, use the part number suffix -1, and for silver, use the part number suffix -5.*

Assembling front barrel

Tools Required:

- #2 Phillips head screwdriver

WARNING: During assembly, the shutter springs can pop out of place. Always wear protective eyewear during this procedure.

1: Stand the top and bottom front barrel castings (6 and 7) upright with the shutter end of the castings toward the bottom.

Note: *The top front barrel casting contains the iris slot.*

2: Slide the bottom divider plate (2), into its place in the bottom casting (7). The dimples on the divider plate must point down.

Note: *The bottom divider plate (2) has four dimples punched into the surface; the top plate (3) has none. The middle divider plates (4) are noticeably thinner-gauge metal than the other two.*

The notches on the divider plates must fit against the flanges in the casting so the plates do not move.

WARNING: Divider plate edges are sharp. Handle with caution.

3: Assemble the divider plates and shutter blades as shown in *Figure 13*. For the shutter blade between the divider (4) and the bottom divider plate (2), pull the blade and plate assembly slightly forward to allow the handle to slip through the slot in the bottom casting, and then fully seat the blade assembly into the casting.

Note: *Make sure no shutter assembly components are under the pattern holder guides.*

4: With both front barrel castings still standing upright, join the two halves, sliding the handle of the top shutter blade through the slot in the top front barrel casting (6).

5: Starting at the bottom of the castings (closest to the shutters), use four PHMS screws (5) to fasten the front barrel casting halves together, as shown. Hand tighten only (25 inch pounds).

Note: *The ends of the two front barrel castings must be even. Adjust as necessary before completely tightening the nuts and screws. Failure to do this could interfere with the barrel rotation.*

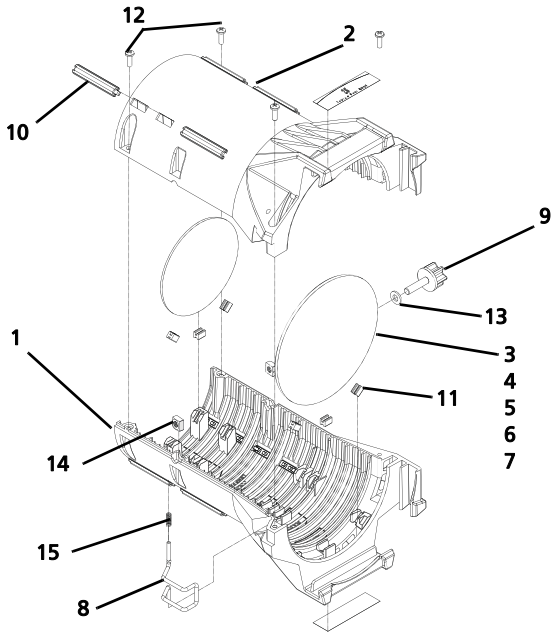
6: Turn the front barrel assembly over so that the narrower end is on your work surface.

7: Install the four shutter springs (14) between the four dimples in the shutter plate and the tabs in the lip of the casting.

Note: *Install the springs at the joints in the castings on either of the tabs at the joint. Once they are installed, the springs at the joints will sit at a slight angle.*

8: Place iris slot cover (5) over the iris slot. Use two screws (11) to secure the cover.

Lens Tubes: 19° to 50°



Note: Lenses are degree specific. Do not interchange lenses (i.e., do not place a 19° lens in a 26° slot).

Figure 14

Reference Number	Part Number	Description	Quantity Required
1	7060A3169	Lens Tube, left casting, Black (rev. J begins 01/06 to present)	1
2	7060A3171	Lens Tube, right casting, Black (rev J. begins 01/06 to present)	1
3	7060A4077	Aspheric lens, 19° (begins 10/98)	1
4	7060A4001	Aspheric lens, 26°	1
5	7060A4020	Meniscus lens (front), 36°	1
6	7060A4021	Bi-convex lens (back), 36°	1
7	7060A4004	Aspheric lens, 50°	1
8	7060A3079	Clip, gel retainer, 90° bend	1
9	7060A4008-01	Knob set with male insert	1
10	7060A4009	Bushing, guide	8
11	7060A4012	Pad, lens support	5
12	HW3154	Screw, Phillips, 8-32 x 5/8 Taptite	4
13	HW5143	Washer, flat fiber	1
14	HW5494	Nut, 1/4-20, 7/16, 7/32, square steel	2
15	HW750	Spring, retainer	1
16	7060A4033	19° Lens tube label, white text (for black fixtures)	1
17	7060A4034	26° Lens tube label, white text (for black fixtures)	1
18	7060A4035	36° Lens tube label, white text (for black fixtures)	1
19	7060A4036	50° Lens tube label, white text (for black fixtures)	1
20	7060A4073	19° Lens tube label, black text (for white fixtures)	1
21	7060A4074	26° Lens tube label, black text (for white fixtures)	1
22	7060A4075	36° Lens tube label, black text (for white fixtures)	1
23	7060A4076	50° Lens tube label, black text (for white fixtures)	1

Note: *Any part that is indicated as "Black" may be also ordered in white or silver. When ordering white, use the part number suffix -1, and for silver, use the part number suffix -5.*

Assembling lens tube: 19° to 50°

Lens Positions

The lens tube casting can accept either original or Enhanced Definition (ED) lenses. The lens positions are color-coded within the lens casting to match the colored dots on the lenses (see *Figure 15*). Only lenses bearing at least one dot of the indicated color should be placed in that position.

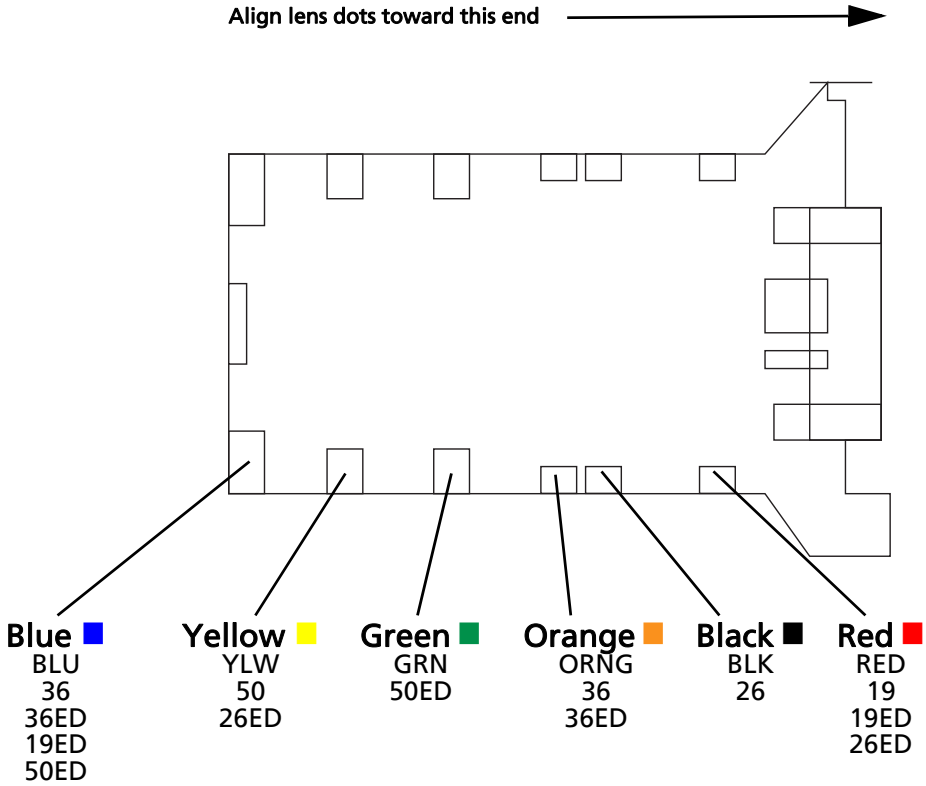
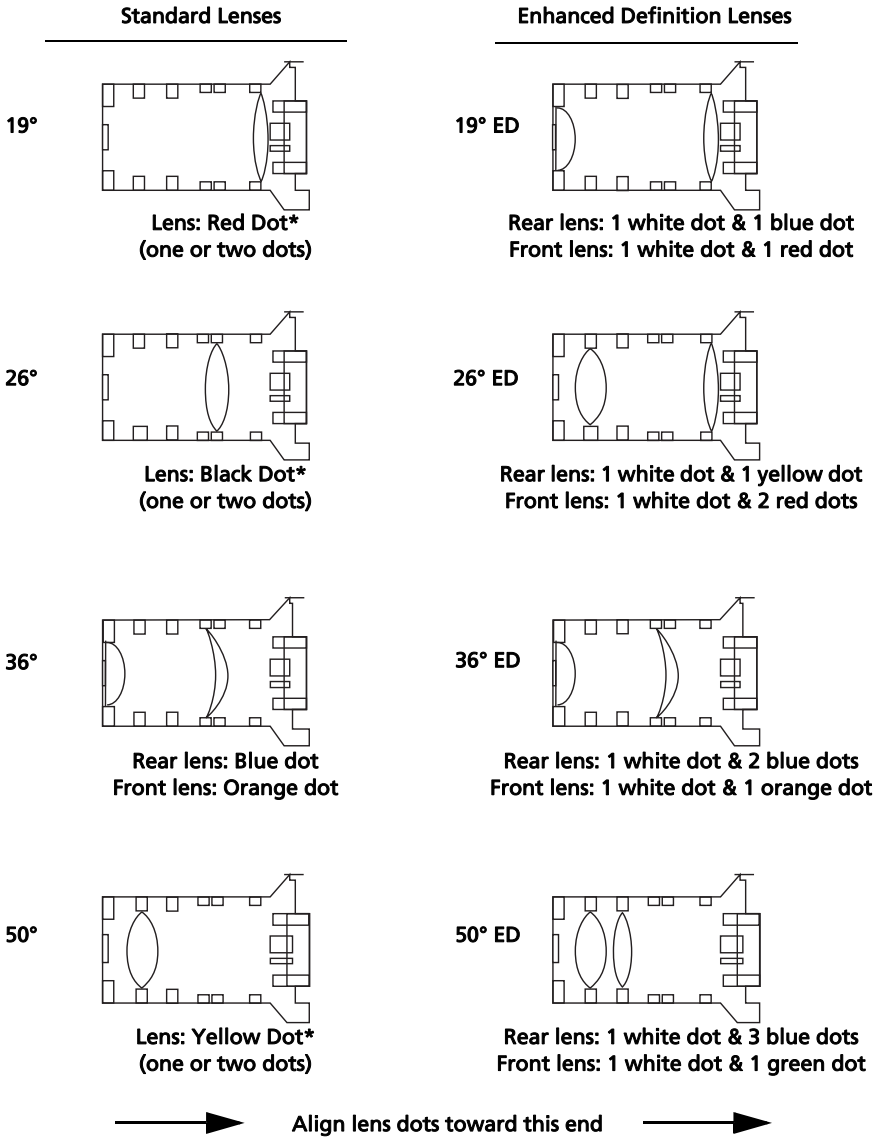


Figure 15

Lens Markings

The lenses are color-coded for identification. Normal lenses have a single dot of color while Enhanced Definition lenses have a combination of white dots and color dots. See *Figure 16* for lens color coding.



*** Some pre-2006 fixtures have lenses with two color code dots. There is no difference between lenses with one dot or two dots.**

Figure 16

For maximum gel life with specific lens tubes, refer to the tables below.

Lens Tube	Soft Focus Back (Lamp "flat")	Sharp Focus (Lamp "cosine")	Soft Focus Forward (Lamp "peak")	Soft Focus Forward (Lamp "cosine")
19°	worse	better	better	not applicable
26°	worse	better	better	not applicable
36°	better	better	worse	not applicable
50°	worse	better	better	not applicable

ED Lens Tube	Soft Focus Back (Lamp "flat"	Sharp Focus (Lamp "cosine")	Soft Focus Forward (Lamp "peak")	Soft Focus Forward (Lamp "cosine")
19°	worse	better	better	not applicable
26°	better	better	worse	not applicable
36°	better	better	worse	not applicable
50°	better	better	worse	not applicable

Assembling the lens tube

Tools Required:

- #2 Phillips head screwdriver

1: Place the left and right lens tubes (1 & 2) face up on your work surface. Align the tubes so the colorframe grooves are to your left.

2: Install lens support pads (11) as required inside both lens tubes. Four pads are required per lens. Insert pads short-side down, as shown in *Figure 14*.

3: Install the 1/4-20 nut (14) in the left lens tube as shown.

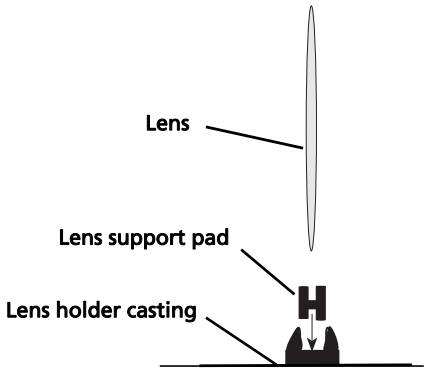


Figure 17

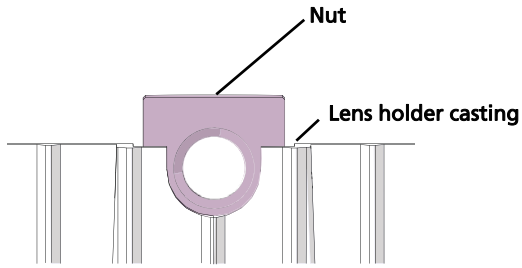


Figure 18

4: Install the short end of the gel retainer clip (8) in the left lens tube (1).

5: Position the clip in the forward, locked position, then install the retainer spring (15) on the clip.

6: Install the required lens (or lenses) (3, 4, 5, 6, 7, or 8) in the correct position as shown on *Figure 15*.

Note: *Install the lens with the painted dot facing the front of the tube. Seat the lens in the support pads so the dot remains visible.*

7: See *Figure 14*. Invert the right lens tube (2) and hold it above the left lens tube (1). Fit the clip (8) and spring (15) into the right lens tube. Gently place the right lens tube onto the left lens tube, making sure that the 1/4-20 nut and retaining clip assembly stay properly seated. Make sure the lens stays straight by looking into the lens holder. The top edge of the lens should seat properly in the support pads.

8: Install the PHMS screws (12) in four locations with nuts (14). The PHMS screw location on the bottom of the colorframe holder is threaded and does not require a nut. Hold the nuts tight against the casting and torque the screws to 25 inch pounds.

9: Install the eight bushing guides (10). Point the narrow tab on the bottom toward the back of the tube; point the square tab toward the front. Squeeze the guides slightly so they bend in the middle then snap into place. Make sure the rounded side of the guide faces outward from the tube.

Cleaning glass lens: 19° to 50° and all EDLT lenses

WARNING: Do not use ammonia-based or other harsh commercial cleaners. Clean lens only as directed.

Commercially available glass cleaning agents should be avoided as they may contain ammonia, other harsh chemical detergents or abrasive agents. These cleaners may damage the glass surface and the Anti-Reflective coatings. Do not immerse or soak the glass in any cleaning solution

Remove dust with a blast of oil-free air or wipe with a clean, lint-free cloth. Isopropyl alcohol, distilled water or a 50%-50% mixture of each can be used to clean the glass surface.

Lens Tube: 10°

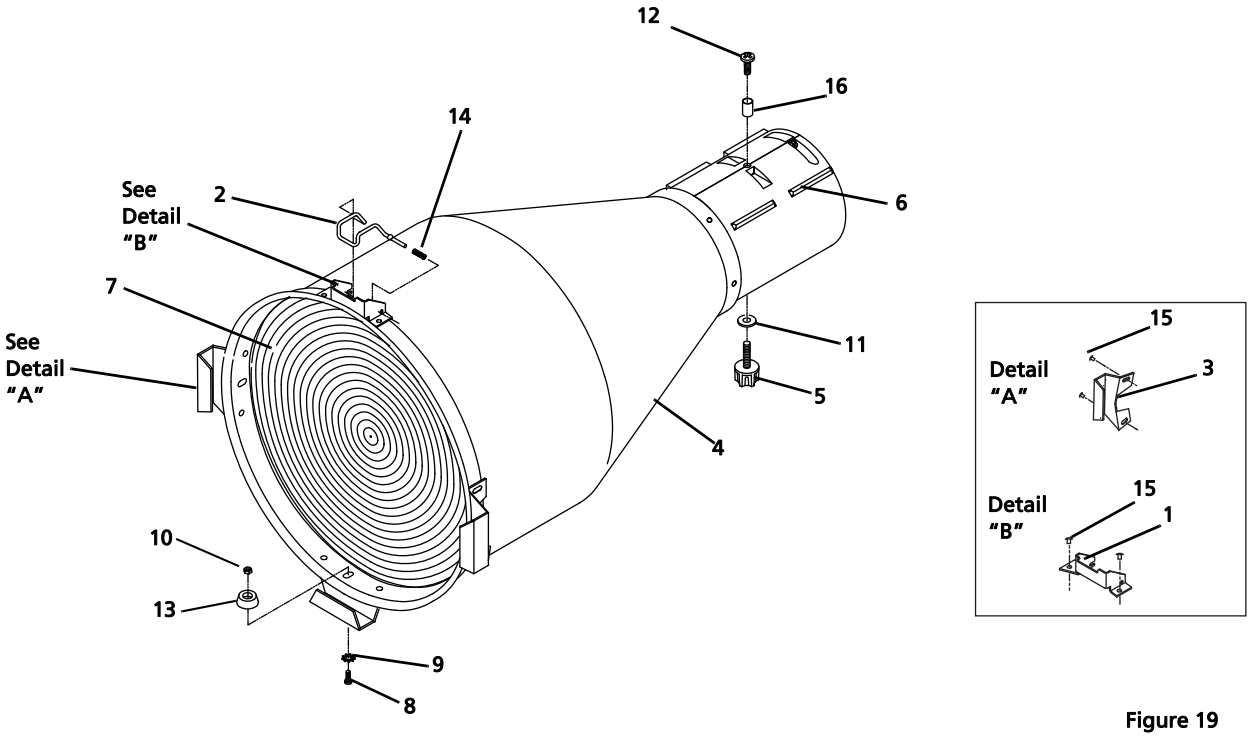


Figure 19

Reference Number	Part Number	Description	Quantity Required
1	7060A3066	Bracket, gel clip	1
2	7060A3079	Clip, gel retainer with 90° bend	1
3	7060A3086	Gel holder, 5° and 10°	3
4	7060A3096	10° lens tube assembly, Black, empty	1
5	7060A4008-01	Knob set with male insert	1
6	7060A4009	Bushing, guide	8
7	7060A4025	Lens, 10°, 10 inch	1
8	HW307	Screw, 8-32 x 36LG, SPHMS, black zinc	4
9	HW3104	Washer, 170 x 381 x 023, black zinc	4
10	HW370	Nut, Ny-lok, 8/32, black zinc	4
11	HW5143	Washer, flat fiber	1
12	HW5197	Screw, 1/4-20 x 5/8", black zinc	1
13	HW6122	Bumper, rubber	4
14	HW750	Spring, retainer	1
15	HW759	Rivet, machined, 125 x 125LG oval HD, ST	8
16	HW5200	Washer, SH, .253 x .281 x .438, black, zinc	1

Note: *Any part that is indicated as "Black" may be also ordered in white or silver. When ordering white, use the part number suffix -1, and for silver, use the part number suffix -5.*

Lens Tube: 5°

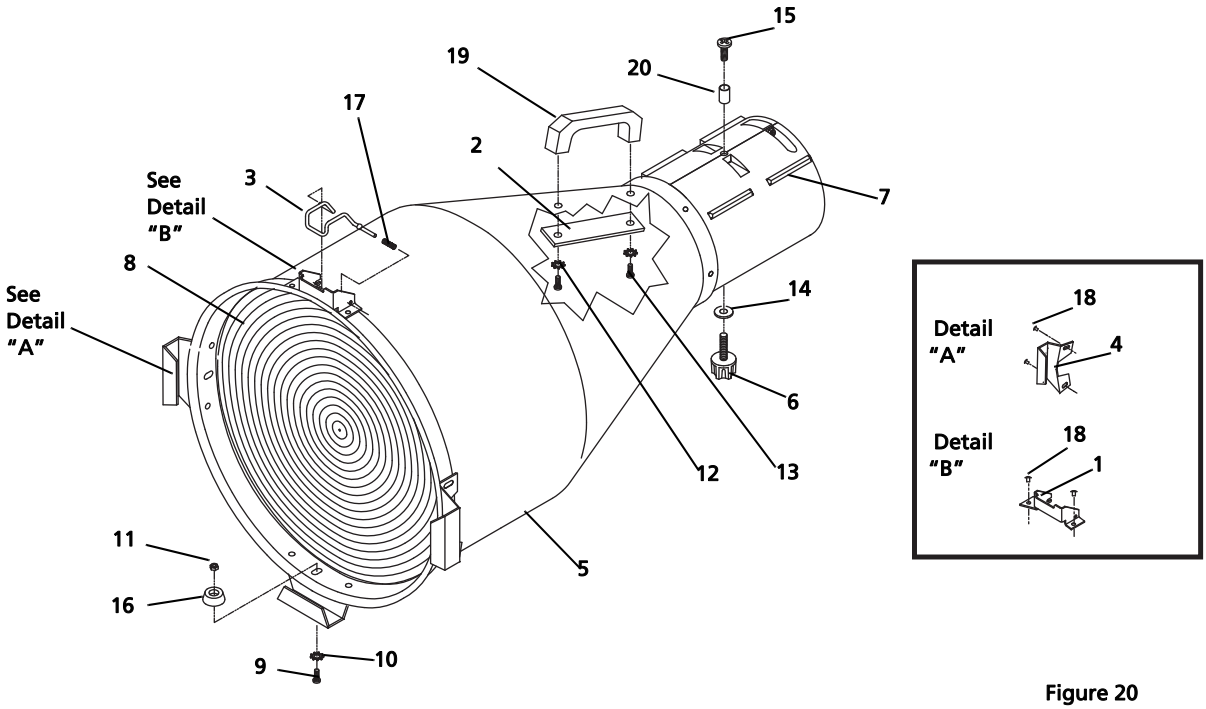


Figure 20

Reference Number	Part Number	Description	Quantity Required
1	7060A3066	Bracket, gel clip	1
2	7060A3073	Handle backing plate	1
3	7060A3079	Clip, gel retainer with 90° bend	1
4	7060A3086	Gel holder, 5° and 10°	3
5	7060A3095	5° lens tube assembly, Black, empty	1
6	7060A4008-01	Knob set with male insert	1
7	7060A4009	Bushing, guide	8
8	7060A4024	Lens, 5°, 12"	4
9	HW307	Screw, 8-32 x 38LG, SPHMS, black zinc	4
10	HW3104	Washer, 170 x 381 x 023, black zinc	4
11	HW370	Nut, Ny-lok, 8/32, black zinc	4
12	HW443	Washer, 195 x 410 x 025, black zinc	2
13	HW467	Screw, 10 x 32 x 1/2" PHTRMS	2
14	HW5143	Washer, flat fiber	1
15	HW5197	Screw, 1/4-20 x 5/8", black zinc	1
16	HW6122	Bumper, recess rubber	4
17	HW750	Spring, retainer	1
18	HW759	Rivet, machined, 125 x 125LG oval HD, ST	8
19	HW8170	Handle, 10-32 inserts	1
20	HW5200	Washer, SH, .253 x .281 x .438, black, zinc	1

Note: *Any part that is indicated as "Black" may be also ordered in white or silver. When ordering white, use the part number suffix -1, and for silver, use the part number suffix -5.*

Assembling lens tubes: 5° and 10°

Tools Required:

- #2 Phillips head screwdriver

1: Place the lens tube assembly (5) on your work surface. Align the assembly so that the colorframe grooves are to your left.

2: If you are assembling a 5° tube, attach the handle (19) as shown in *Figure 20*, using the screws (13), washers (12) and backing plate (2).

3: Install the required lens as shown in *Figure 19* or *Figure 20* in the tube using required bumpers (16), screws (9), nuts (11), and washers (10). The side of the lens with the fresnel grooves should face the front of the tube. The smooth side should face the rear.

Cleaning polymer lens: 5° and 10°

WARNING: Handle polymer lenses by their edges only. Never rub anything dry on a polymer lens. Do not use glass and window cleaners on the lens. This will damage the lens.

Remove dust with a blast of oil-free air. If this is not sufficient, follow the instructions below.

1: Dip the lens in a clean isopropyl alcohol/water mixture (9 parts water to 1 part isopropyl alcohol).

2: Use a soft moistened nylon bristle brush to wash the lens' smooth side in a linear (non-circular) motion.

3: Use the same brush to lightly wash the lens' ridged side by following its ridges.

4: Dip the lens in a clean isopropyl alcohol/water mixture (9 parts water to 1 part isopropyl alcohol).

5: Use air gun to dry the smooth surface. Use air gun to dry the ridged surface. Use air stream to move the liquid away from you. Using this method, remove as much liquid as possible. Inspect the lens for dirt. If necessary, repeat the entire process.

Lens Tube: 14°

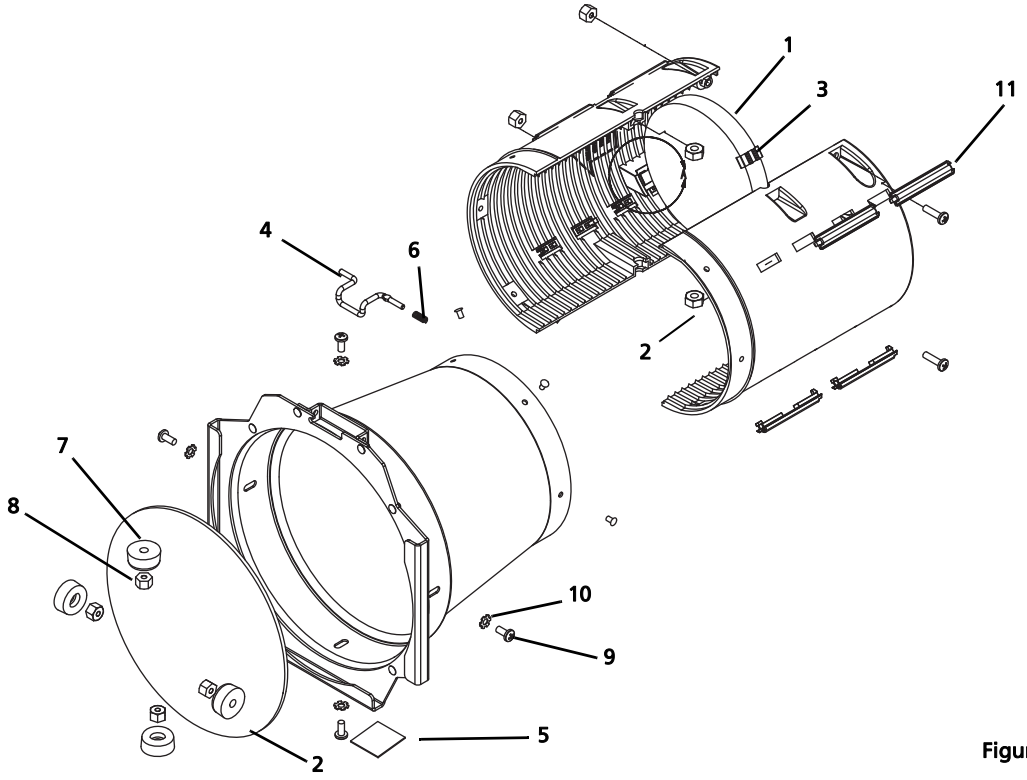


Figure 21

Reference Number	Part Number	Description	Quantity Required
1	7060A4124	14° Lens , Rear	1
2	7060A4125	14° Lens, Front	1
3	7060A4142	Lens Pad	4
4	7060A3180	Gel clip, lens tube (common to the S4, 14°, 70°, and 90° lens tubs)	1
5	7060A4143	Label, 14°	1
6	HW750	Spring retainer (common to the S4 14°, 70°, and 90° lens tubes)	1
7	HW6122	Bumper, rubber	4
8	HW370	Nut, 8-32	4
9	HW307	Screw, Phillips, 8-32 x 3/8, PHMS	4
10	HW3104	Washer, #8	4
11	7060A4009	Bushing guide	8

For maximum gel life with specific lens tubes, refer to the table below.

Lens Tube	Soft Focus Back (Lamp "flat")	Sharp Focus (Lamp "cosine")	Soft Focus Forward (Lamp "peak")	Soft Focus Forward (Lamp "cosine")
14°	worse	good	best	good

Note: *The 14° lens tube casting assembly is not field repairable. Please contact ETC Customer Service for repair information.*

Assembling lens tube: 14°

Lens Markings and Positions

The lenses are color-coded for identification. The rear lens has two brown dots and the front lens has one brown dot. The lens positions are color-coded within the lens casting to match the colored dots on the lenses (see *Figure 22*).

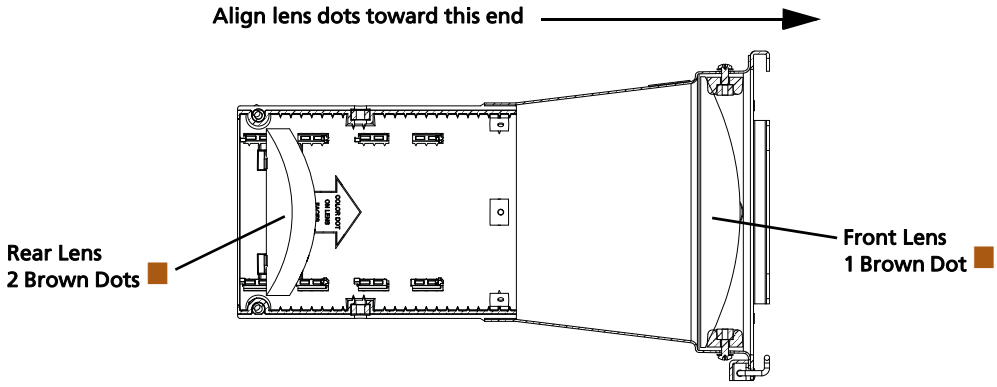


Figure 22

Assembling the lens tube

Tools Required:

- #2 Phillips head screwdriver

1: Place lens tube assembly face up on your work surface.

2: Install Install the eight bushing guides (11). Point the narrow tab on the bottom toward the back of housing; point the square tab toward the front. Squeeze the guides slightly so they end inthe middle, then snap into place. Make sure the rounded side of the guide faces outward from the tube.

3: Install the front lens (2) in the tube using required bumpers (7), screws (9), nuts (8), and washers (10) as shown in *Figure 21*. Colored dot on lens should face towards the front.

Cleaning glass lenses: 14°

WARNING: Do not use ammonia-based or other harsh commercial cleaners. Clean lens only as directed.

Commercially available glass cleaning agents should be avoided as they may contain ammonia, other harsh chemical detergents or abrasive agents. These cleaners may damage the glass surface and the Anti-Reflective coatings. Do not immerse or soak the glass in any cleaning solution.

1: Remove dust with a blast of oil-free air or wipe with a clean, lint-free cloth. Isopropyl alcohol, distilled water or a 50%-50% mixture of each can be used to clean the glass surface.

2: If necessary, remove the front lens to access the front surface of the rear lens for cleaning. Remove the four screws (9), nuts (8), washers (10) and bumpers (7) and gently left out front lens as shown in *Figure 21*. Reassemble in reverse order.

Lens Tube: 70°

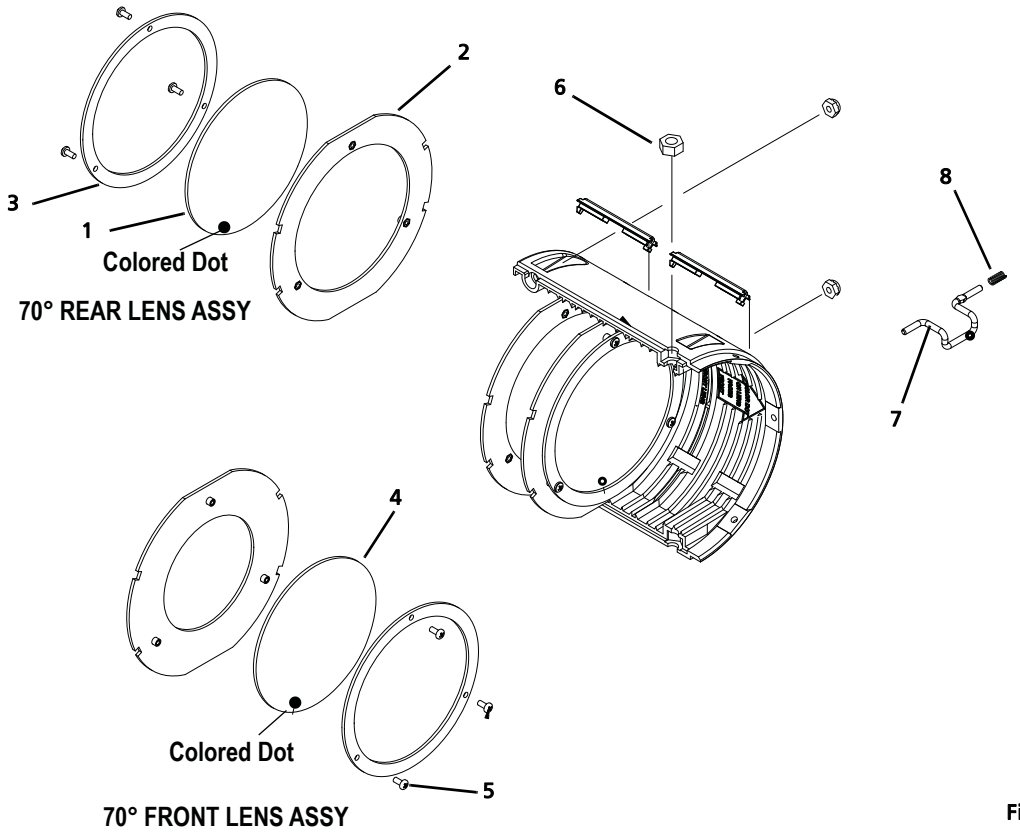


Figure 23

Reference Number	Part Number	Description	Quantity Required
1	7060A4134	Rear lens, 70°	1
2	7060A4135	Front lens, 70°	1
3	7060A3209	Retaining ring	2
4	7060A4144	Label, 70°	1
5	HW1119	Screw, Phillips, 4-40 x 1/4, SEMS, PHMS	6
6	HW534	Nut, Hex, 1/4-20	2
7	7060A3180	Gel clip, lens tube (common to the S4, 14°, 70°, and 90° lens tube)	1
8	HW750	Spring retainer (common to the S4 14°, 70°, and 90° lens tubes)	1

For maximum gel life, refer to the table below:

Lens Tube	Soft Focus Back (Lamp "flat")	Sharp Focus (Lamp "cosine")	Soft Focus Forward (Lamp "peak")	Soft Focus Forward (Lamp "cosine")
70°	best	good	worse	good

Note: *The 70° lens tube casting assembly is not field repairable. Please contact ETC Customer Service for repair information.*

Assembling lens tube: 70°

Lens Markings and Positions

The lenses are color-coded for identification. The rear lens has two green dots and the front lens has one green dot. The lens positions are color-coded within the lens tube casting to match the colored dots on the lenses (see *Figure 24*).

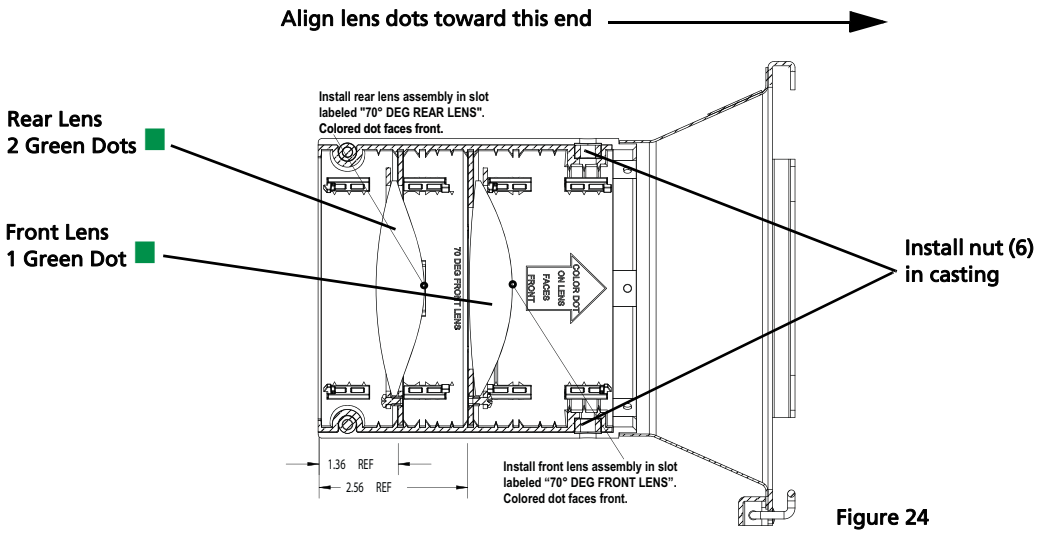


Figure 24

Assembling the lens tube

Tools Required:

- #2 Phillips head screwdriver

1: See *Figure 23*. Place rear lens (1) into mounting plate (2) with paint dots facing forward. Center the lens in the mounting plate and secure with the retaining ring (3) and screws (6). Lightly tighten all three screws while maintaining lens position, then tighten screws securely.

2: See *Figure 23*. Place front lens (4) into mounting plate (5) with paint dots facing forward. Center the lens in the mounting plate and secure with the retaining ring (3) and screws (6). Lightly tighten all three screws while maintaining lens position, then tighten screws securely.

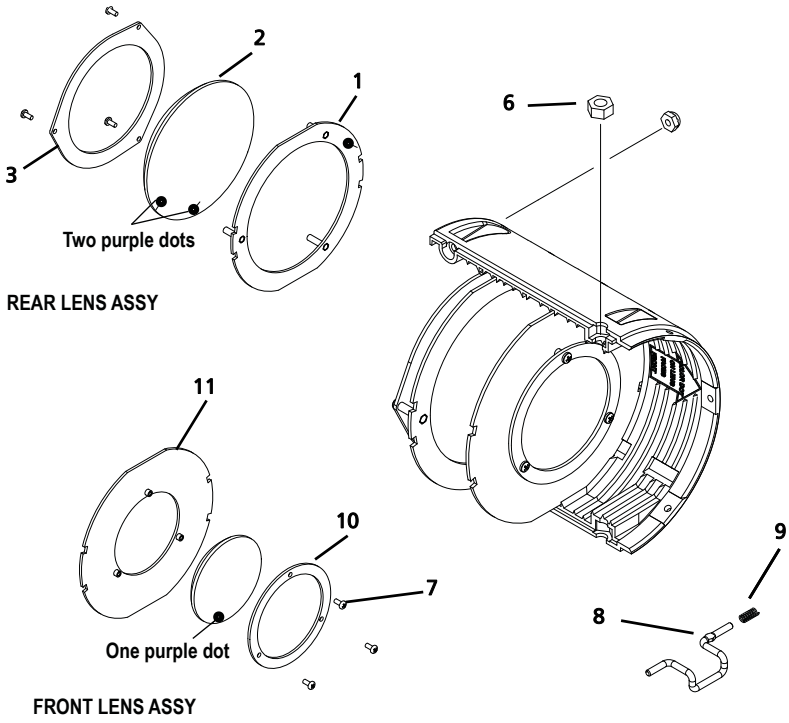
Cleaning glass lens: 70°

WARNING: Do not use ammonia-based or other harsh commercial cleaners. Clean lens only as directed.

Commercially available glass cleaning agents should be avoided as they may contain ammonia, other harsh chemical detergents or abrasive agents. These cleaners may damage the glass surface and the Anti-Reflective coatings. Do not immerse or soak the glass in any cleaning solution.

Remove dust with a blast of oil-free air or wipe with a clean, lint-free cloth. Isopropyl alcohol, distilled water or a 50%-50% mixture of each can be used to clean the glass surface.

Lens Tube: 90°



Label Orientation **Figure 25**

Reference Number	Part Number	Description	Quantity Required
1	7060A4136	Rear lens, 90°	1
2	7060A4137	Front, lens, 90°	1
3	7060A3182	Retaining ring, 90°, rear lens	1
4	7060A3190	Retaining ring, 90°, front lens	1
5	7060A4145	Label, 90°	1
6	HW534	Nut, Hex, 1/4-20	2
7	HW1119	Screw, Phillips, 4-40 x. 1/4, SEMS, PHMS	6
8	7060A3180	Gel clip, lens tube (common to the S4 14°, 70°, and 90° lens tubes)	1
9	HW750	Spring retainer (common to the S4 14°, 70°, and 90° lens tubes)	1
10	7060A3184	Retaining ring, 90°, front lens	1
11	7060A3183	Mounting plate, 90°, front lens	1

For maximum gel life, refer to the table below:

Lens Tube	Soft Focus Back (Lamp "flat")	Sharp Focus (Lamp "cosine")	Soft Focus Forward (Lamp "peak")	Soft Focus Forward (Lamp "cosine")
90°	best	good	good	worse

Note: *The 90° lens tube casting assembly is not field repairable. Please contact ETC Customer Service for repair information.*

Assembling lens tube: 90°

Lens Markings and Positions

The 90° lenses are color-coded for identification. The rear lens has two purple dots and the front lens has one purple dot. The lens positions are color-coded within the lens casting to match the colored dots on the lenses (see *Figure 26*).

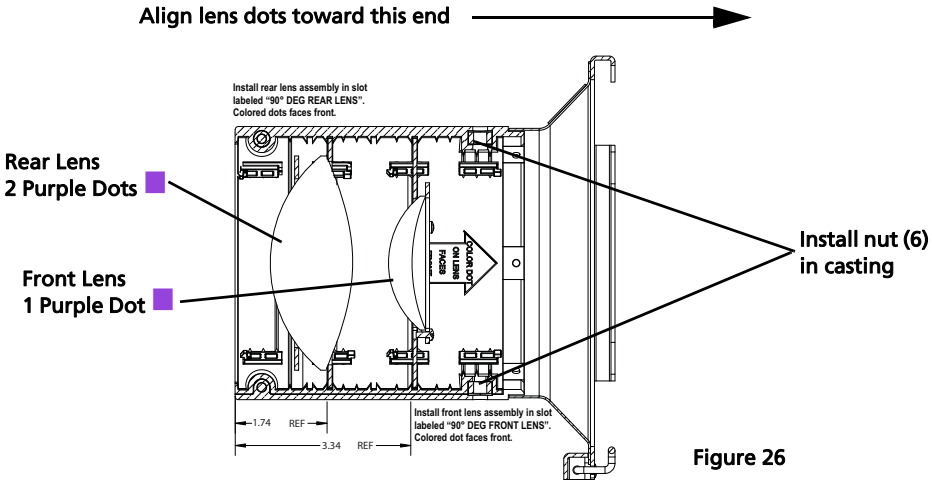


Figure 26

Assembling the lens tube

Tools Required:

- #2 Phillips head screwdriver

1: See *Figure 25*. Place rear lens (2) into mounting plate (1) with paint dots facing forward. Center the lens in the mounting plate and secure with the retaining ring (3) and screws (7). Lightly tighten all three screws while maintaining lens position, then tighten screws securely.

2: See *Figure 25*. Place front lens (2) into mounting plate (11) with paint dots facing forward. Center the lens in the mounting plate and secure with the retaining ring (10) and screws (7). Lightly tighten all three screws while maintaining lens position, then tighten screws securely.

Cleaning glass lens: 90°

WARNING: Do not use ammonia-based or other harsh commercial cleaners. Clean lens and reflector only as directed.

Commercially available glass cleaning agents should be avoided as they may contain ammonia, other harsh chemical detergents or abrasive agents. These cleaners may damage the glass surface and the Anti-Reflective coatings. Do not immerse or soak the glass in any cleaning solution.

Remove dust with a blast of oil-free air or wipe with a clean, lint-free cloth. Isopropyl alcohol, distilled water or a 50%-50% mixture of each can be used to clean the glass surface.



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